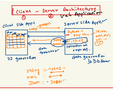
HTML, CSS, JavaScript, Bootstrap, React, Angular → Front-End Web Development

Python, Flask → Server-side Web Development

Front-End Web Developer Roles:

1) Creating the user interface for the web application
2) Bringing the application alive
3) Making the user view the application
4) Connecting the back-end and the data set to meaningful visual

Client - Server Architecture / Web Application



Client Side Application: Front-End Web

- HTML
- CSS
- JavaScript
- React
- Angular

Server Side Application: Back-End Web

- Java
- PHP
- .NET (C#, Asp.net, MVC, C++)
- Python → Flask, ...

Client Side Application:
Front-End Web Application!

HTML (Foundation) — CSS (Style) — JavaScript (Functionality)

interior, colour → function, layout

+ light → interaction, life →

Web Page(s)
↓
Web site application.

HTML → Static Website.
↓
interactive page(s) but they cannot interact with server or database

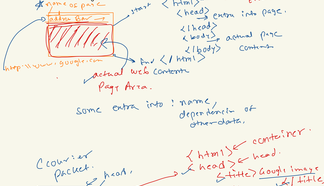
Web Page(s)

Dynamic Website.

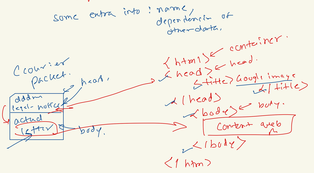
- interactive pages
- The content gets loaded may vary as per requirement.
- We can interact with database

HTML → HyperText Markup language.
↓ hyperlink
↓ annotated
<>

HTML Document
tags :- HTML Elements



some extra info : name, dependency of other data.



name of tag → HTML Tags.

<h1> This is a header </h1>

Opening Tags — content — closing of tag.

HTML Element.

html elements / tags.

- <img /> ← self closing tag
-
- <input />
- <hr />

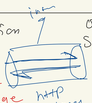
<h1 border="1">Hello</h1>

name of tag — attributes — content — closing tag.

Opening tag

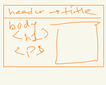
OS: Windows
Client Side Application

1) Browser app.
   1) Google : Chrome
   2) Microsoft : MS Edge



OS: Windows → Server
Server side Application.
base Appln.

host app → Python code
store data → Db

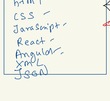


HTTP Protocol

+91 : India
+633 : USA

+63 575-232-61 +91 # 8727000228

Browser | Client



http://www.bcc.com

Server side
emit data

- C++
- PHP
- Java
- .Net
- Python

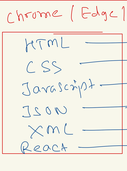
emit O/P

HTTP + html, CSS, JS, JSON

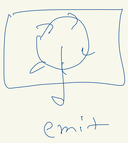
emit data in such format, which browser application can understand it.

emitted data / server side generated data
↓
html output
CSS op
JS op
img / pdf / text / plain

Browser chrome | Edge | Firefox



Web Server



emit output

Tomcat / Apache / IIS / Kestrel

= html, css, JS, JSON, xml, text, img output

Web server [live server]

80

5500

http://www.bvs.com/index.html

Browser

Client

HTTP Request Packet

http://127.0.0.1:5500/index.html

loop back address

client m/c = server m/c

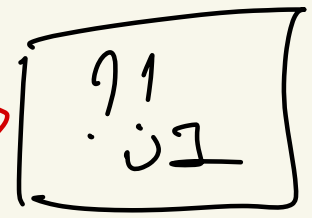
port

no live server

Index.html

```
<html>
  <head>
    <title> Index </title>
  </head>
  <body>
    <h1> Heading </h1>
    <p> <span> Hello </span> </p>
  </body>
</html>
```

HTTP Response Packet



Resp

1234

5678

IP: Unique ID

IP: UI.

Req.

IP:

Who

when

what

what expects.

Browser

URL: http://www.brs.com/index.html

→ http method.

Packet sent to Serv

Live Server

5500

index.html

<html>
<head>
<body
<p
</html>

unloads → Request packet.

URL :- brs.com/index.html

what dir = path of your app

find dir first.

if found.

if not found

Bad Req

404

dir

index.html.

generate HTTP Response Packet.

head
→ To
→ from
URL : "brs/index.html"
lang : en-us
method : GET
brows IP
Date

body

http Request Packet

Browser

UI

Heading

some Text

load html parser module

Sending http resp

unload Resp. Packet

→ head : "text/html"

→ body → index.html content.

To: Browser IP
from: server IP
Content-Type:
"text/html" → html

Body
<html>
Content of page
</html>

application/png → img.png

→ application/json → json data

text/plain → only plain text

top to bottom

one line at a time

<html>

<head>

<title> Home </title>

</head>

<body>

 Some Text

</body>

</html>

HTML Parser

Home

Some Text

Browser UI

Standard HTML ← old HTML

html 1.0

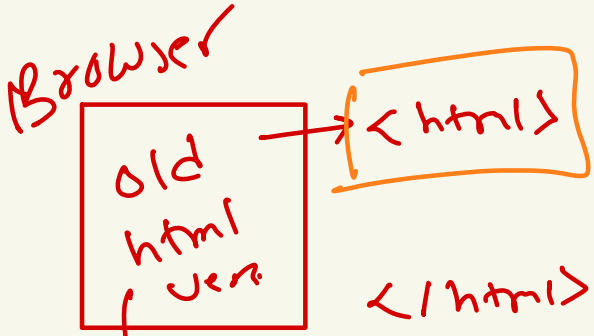
html 2.0

html 3.0

html 4.0

html versions → W3C Organization.

Browser

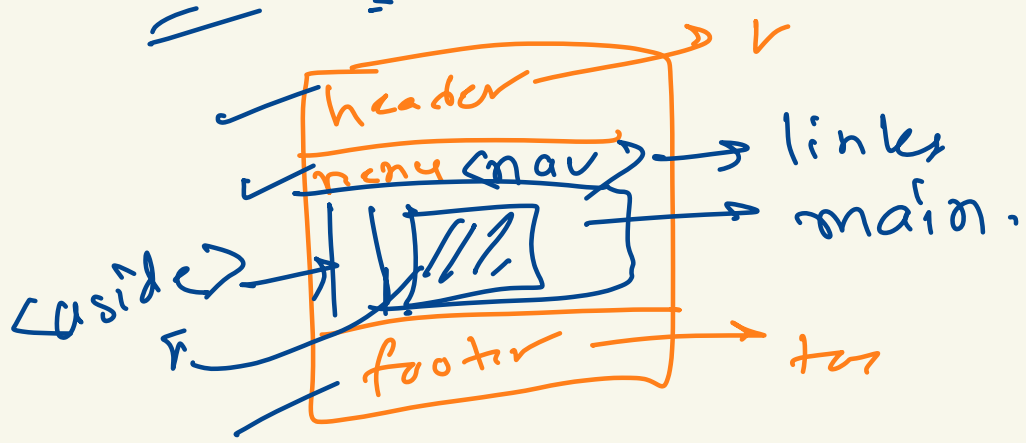


loads html parser which can interprete older version of html documents.

html, head, title, body, h1 – h6, p, span, div, b, table, tr, td, th ....

HTML 5

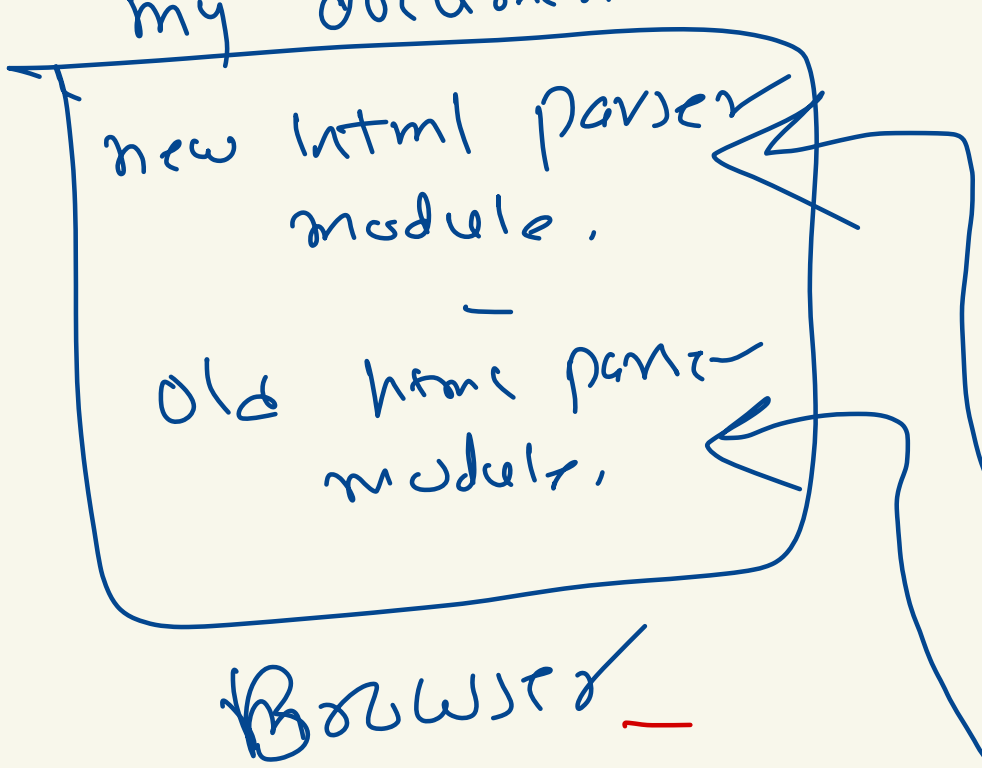
thead, tbody, em, strong,
header, footer, main, nav → main content
aside, fig,
svg, canvas



HTML Elements :- old html Tags

HTML Elements :- Semantic Element.
with meaning

my document uses latest html 5



Respon body.
text / html
unload.
<!Doctype html>
HTML 5
unload
<html>

HTML 5

- Semantic Elements.
  → These are HTML elements that describe the meaning of the element to the browser and the user

- Drag and Drop.
  This is used to drag and drop elements from one position to another.

- Persistant Local storage.
  This is localised storage used to remove dependency on third-party plugins.

- Canvas <canvas> <svg> ~vector graphics
  This
  This is a 2D drawing surface using JS.

- Geolocation
  Using this, visitors can choose to share their location with the websites requesting it.